

Title

Nested Process Evaluation of a Theory-Informed Behaviour Change Intervention Embedded Within Cardiac Rehabilitation Services

Abstract

Cardiovascular disease recovery requires sustained engagement with structured exercise and lifestyle modification programmes. This process evaluation will be nested within a cluster-randomised trial comparing standard cardiac rehabilitation versus rehabilitation augmented with explicit behaviour change coaching. Participants in both arms will attend supervised exercise sessions and education classes. The intervention arm will receive additional structured coaching incorporating goal-setting, self-monitoring, and motivational strategies grounded in established behaviour change theory. Data collection will include semi-structured interviews with participants from both conditions, focus group discussions with intervention participants, attendance records, and participant reflective diaries. Analysis will explore perceived barriers and facilitators to maintaining physical activity, the influence of social support networks, and contextual factors shaping programme engagement. Qualitative findings will be analysed thematically to identify mechanisms through which behaviour change support operates. This process evaluation will clarify which elements of standard rehabilitation and added coaching contribute most meaningfully to sustained behaviour change.